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1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a Consultation Charge System. Design a C++ program for hospital management

Task:

A. Use function overloading to calculate charge for general consultation and specialist consultation.

The screenshot shows the SIMATS Online Programming Lab interface. The code is as follows:

```
#include <iostream>
using namespace std;

float calculateCharge(int hours);
float calculateCharge(int hours, float specialistfee);
void modifyBill(float &bill, float serviceCharge = 100);

int main()
{
    int choice;
    int hours;
    float specialistfee;
    float totalBill;
    string patientType;

    cout << "Hospital Consultation Charge System" << endl;
    cout << "Enter Patient Type:" << endl;
    cout << "1. General Consultation" << endl;
    cout << "2. Specialist Consultation" << endl;
    cout << "Enter Choice: ";
```

The output shows the program execution:

```
Hospital Consultation Charge System
Enter Patient Type:
1. General Consultation
2. Specialist Consultation
Enter Choice: Enter Consultation Hours:
Patient Type: General Patient
Total Consultation Charge: Rs. 1100
```

b. Use call by reference to modify the final bill amount.

The screenshot shows the SIMATS Online Programming Lab interface. The code is as follows:

```
#include <iostream>
using namespace std;

float calculateCharge(int hours);
float calculateCharge(int hours, float specialistfee);
void modifyBill(float &bill, float serviceCharge = 100);

int main()
{
    int choice;
    int hours;
    float specialistfee;
    float totalBill;
    string patientType;

    cout << "Hospital Consultation Charge System" << endl;
    cout << "Enter Patient Type:" << endl;
    cout << "1. General Consultation" << endl;
    cout << "2. Specialist Consultation" << endl;
    cout << "Enter Choice: ";
```

The output shows the program execution:

```
Hospital Consultation Charge System
Enter Patient Type:
1. General Consultation
2. Specialist Consultation
Enter Choice: Enter Consultation Hours:
Patient Type: General Patient
Total Consultation Charge: Rs. 1600
```

c. Use default arguments to add a standard service charge if needed.

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```

1 #include <iostream>
2 using namespace std;
3
4 float calculateCharge(int hours);
5 float calculateCharge(int hours, float specialistFee);
6 void modifyBill(float &bill, float serviceCharge = 100);
7
8 int main()
9 {
10     int choice;
11     int hours;
12     float specialistFee;
13     float totalBill;
14     string patientType;
15
16     cout << "Hospital Consultation Charge System" << endl;
17
18     cout << "Enter Patient Type:" << endl;
19     cout << "1. General Consultation" << endl;
20     cout << "2. Specialist Consultation" << endl;
21     cout << "Enter Choice: ";

```

```

2
2
1000

```

```

Hospital Consultation Charge System
Enter Patient Type:
1. General Consultation
2. Specialist Consultation
Enter Choice: Enter Consultation Hours: Enter Specialist Fee:
Patient Type: Specialist Patient
Total Consultation Charge: Rs. 2700

```

d. Use function prototyping for all calculation functions.

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```

1 #include <iostream>
2 using namespace std;
3
4 float calculateCharge(int hours);
5 float calculateCharge(int hours, float specialistFee);
6 void modifyBill(float &bill, float serviceCharge = 100);
7
8 int main()
9 {
10     int choice;
11     int hours;
12     float specialistFee;
13     float totalBill;
14     string patientType;
15
16     cout << "Hospital Consultation Charge System" << endl;
17
18     cout << "Enter Patient Type:" << endl;
19     cout << "1. General Consultation" << endl;
20     cout << "2. Specialist Consultation" << endl;
21     cout << "Enter Choice: ";

```

```

2
5
2000

```

```

Hospital Consultation Charge System
Enter Patient Type:
1. General Consultation
2. Specialist Consultation
Enter Choice: Enter Consultation Hours: Enter Specialist Fee:
Patient Type: Specialist Patient
Total Consultation Charge: Rs. 6100

```

e. Display the patient type and total consultation charge.(BL3)

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```

1 #include <iostream>
2 using namespace std;
3
4 float calculateCharge(int hours);
5 float calculateCharge(int hours, float specialistFee);
6 void modifyBill(float &bill, float serviceCharge = 100);
7
8 int main()
9 {
10     int choice;
11     int hours;
12     float specialistFee;
13     float totalBill;
14     string patientType;
15
16     cout << "Hospital Consultation Charge System" << endl;
17
18     cout << "Enter Patient Type:" << endl;
19     cout << "1. General Consultation" << endl;
20     cout << "2. Specialist Consultation" << endl;
21     cout << "Enter Choice: ";

```

```

3
2

```

```

Hospital Consultation Charge System
Enter Patient Type:
1. General Consultation
2. Specialist Consultation
Enter Choice: Enter Consultation Hours: Invalid Choice

```

2. Design a C++ program Fitness Centre Membership System. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Task:

- Create a class Member with attributes: Member ID, Name, Plan Type, and Fee.

SIMATS Online Programming Lab
C++ PROGRAMMING

1
104
Kiran
Basic

```
#include <iostream>
using namespace std;

class Member
{
private:
    int memberId;
    string name;
    string planType;
    float fee;
public:
    static int totalMembers;
    void getData()
    {
        cout << "Enter Member ID: ";
        cin >> memberId;

        cout << "Enter Member Name: ";
        cin >> name;
    }
};

int main()
{
    Member m1;
    m1.getData();
    cout << endl;
    m1.display();
    return 0;
}
```

OUTPUT

```
Enter Member ID: Enter Member Name: Enter Plan (Basic/Premium/VIP):
----- Membership Details -----
Member ID : 104
Member Name : Kiran
Plan Type : Basic
Membership Fee : Rs. 1000
Total Active Members: 1
```

b) Implement member functions to assign plans, calculate fees, and display membership details.

SIMATS Online Programming Lab
C++ PROGRAMMING

1
102
priya
Premium

```
#include <iostream>
using namespace std;

class Member
{
private:
    int memberId;
    string name;
    string planType;
    float fee;
public:
    static int totalMembers;
    void getData()
    {
        cout << "Enter Member ID: ";
        cin >> memberId;

        cout << "Enter Member Name: ";
        cin >> name;
    }
};

int main()
{
    Member m1;
    m1.getData();
    cout << endl;
    m1.display();
    return 0;
}
```

OUTPUT

```
Enter Member ID: Enter Member Name: Enter Plan (Basic/Premium/VIP):
----- Membership Details -----
Member ID : 102
Member Name : priya
Plan Type : Premium
Membership Fee : Rs. 2000
Total Active Members: 1
```

c) Use an array of objects to manage multiple members.

SIMATS Online Programming Lab
C++ PROGRAMMING

1
103
noki
VIP

```
#include <iostream>
using namespace std;

class Member
{
private:
    int memberId;
    string name;
    string planType;
    float fee;
public:
    static int totalMembers;
    void getData()
    {
        cout << "Enter Member ID: ";
        cin >> memberId;

        cout << "Enter Member Name: ";
        cin >> name;
    }
};

int main()
{
    Member m1;
    m1.getData();
    cout << endl;
    m1.display();
    return 0;
}
```

OUTPUT

```
Enter Member ID: Enter Member Name: Enter Plan (Basic/Premium/VIP):
----- Membership Details -----
Member ID : 103
Member Name : noki
Plan Type : VIP
Membership Fee : Rs. 3000
Total Active Members: 1
```

d) Add validation for valid membership plans.

SIMATS Online Programming Lab
C++ PROGRAMMING

1
101
rahul
Invalid

```
#include <iostream>
using namespace std;

class Member
{
private:
    int memberId;
    string name;
    string planType;
    float fee;
public:
    static int totalMembers;
    void getData()
    {
        cout << "Enter Member ID: ";
        cin >> memberId;

        cout << "Enter Member Name: ";
        cin >> name;
    }
};

int main()
{
    Member m1;
    m1.getData();
    cout << endl;
    m1.display();
    return 0;
}
```

OUTPUT

```
----- Membership Details -----
Member ID : 101
Member Name : rahul
Plan Type : Invalid
Membership Fee : Rs. 0
Total Active Members: 0
```

e) Include a static member to track total active members.

```

1 #include <iostream>
2 using namespace std;
3
4 class Member
5 {
6 private:
7     int memberID;
8     string name;
9     string planType;
10    float fee;
11
12 public:
13    static int totalMembers;
14
15    void getData()
16    {
17        cout << "Enter Member ID: ";
18        cin >> memberID;
19
20        cout << "Enter Member Name: ";
21        cin >> name;
22    }
23
24 OUTPUT
25
26 Member ID : 104
27 Member Name : Kiran
28 Plan Type : Basic
29 Membership Fee : Rs. 1000
30
31 Member ID : 105
32 Member Name : Bob
33 Plan Type : VIP

```

f) Evaluate how your design supports easy modification of pricing plans.

```

1 #include <iostream>
2 using namespace std;
3
4 class Member
5 {
6 private:
7     int memberID;
8     string name;
9     string planType;
10    float fee;
11
12 public:
13    static int totalMembers;
14
15    void getData()
16    {
17        cout << "Enter Member ID: ";
18        cin >> memberID;
19
20        cout << "Enter Member Name: ";
21        cin >> name;
22    }
23
24 OUTPUT
25
26 Plan Type : Premium
27 Membership Fee : Rs. 2000
28
29 Member ID : 107
30 Member Name : prem
31 Plan Type : Invalid
32 Membership Fee : Rs. 0
33
34 Total Active Members: 3

```

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

Task:

A. Use function prototyping to declare EMI-related functions.

```

1 #include <iostream>
2 using namespace std;
3
4 float calculateEMI(float loanAmount, int years, float rate = 7.5);
5 inline float calculateInterest(float amount, float rate, int years);
6 void updatePayable(float &amount, float subsidy);
7
8 int main()
9 {
10    float loanAmount;
11    float rate;
12    float emi;
13    float interest;
14    float payableAmount;
15    float subsidy;
16    int years;
17
18    cout << "Loan Repayment System" << endl;
19
20    cout << "Enter Loan Amount: ";
21    cin >> loanAmount;
22
23 OUTPUT
24
25 Loan Repayment System
26 Enter Loan Amount: Enter Loan Period (Years): Enter Interest Rate (Enter 0 for
27 default): Enter Subsidy Amount:
28 Loan Amount: Rs. 100000
29 Interest Amount: Rs. 37500
30 EMI Amount: Rs. 635.126
31 Final Payable Amount: Rs. 132500

```

b. Use default arguments for default interest rate where applicable.

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```

1 #include <iostream>
2 using namespace std;
3
4 float calculateEMI(float loanAmount, int years, float rate = 7.5);
5 inline float calculateInterest(float amount, float rate, int years);
6 void updatePayable(float &amount, float subsidy);
7
8 int main()
9 {
10     float loanAmount;
11     float rate;
12     float emi;
13     float interest;
14     float payableAmount;
15     float subsidy;
16     int years;
17
18     cout << "Loan Repayment System" << endl;
19
20     cout << "Enter Loan Amount: ";
21     cin >> loanAmount;

```

200000
5
8
10000

OUTPUT

Loan Repayment System
Enter Loan Amount: Enter Loan Period (Years): Enter Interest Rate (Enter 0 for default): Enter Subsidy Amount:
Loan Amount: Rs. 200000
Interest Amount: Rs. 80000
EMI Amount: Rs. 1355.78
Final Payable Amount: Rs. 270000

c. Apply call by reference to update payable amount after subsidy.

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1924231018A

```

1 #include <iostream>
2 using namespace std;
3
4 float calculateEMI(float loanAmount, int years, float rate = 7.5);
5 inline float calculateInterest(float amount, float rate, int years);
6 void updatePayable(float &amount, float subsidy);
7
8 int main()
9 {
10     float loanAmount;
11     float rate;
12     float emi;
13     float interest;
14     float payableAmount;
15     float subsidy;
16     int years;
17
18     cout << "Loan Repayment System" << endl;
19
20     cout << "Enter Loan Amount: ";
21     cin >> loanAmount;

```

50000
2
6
2000

OUTPUT

Loan Repayment System
Enter Loan Amount: Enter Loan Period (Years): Enter Interest Rate (Enter 0 for default): Enter Subsidy Amount:
Loan Amount: Rs. 50000
Interest Amount: Rs. 6000
EMI Amount: Rs. 260.613
Final Payable Amount: Rs. 54000

d. Use inline functions for simple interest calculations.

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```

1 #include <iostream>
2 using namespace std;
3
4 float calculateEMI(float loanAmount, int years, float rate = 7.5);
5 inline float calculateInterest(float amount, float rate, int years);
6 void updatePayable(float &amount, float subsidy);
7
8 int main()
9 {
10     float loanAmount;
11     float rate;
12     float emi;
13     float interest;
14     float payableAmount;
15     float subsidy;
16     int years;
17
18     cout << "Loan Repayment System" << endl;
19
20     cout << "Enter Loan Amount: ";
21     cin >> loanAmount;

```

50000
10
9
25000

OUTPUT

Loan Repayment System
Enter Loan Amount: Enter Loan Period (Years): Enter Interest Rate (Enter 0 for default): Enter Subsidy Amount:
Loan Amount: Rs. 50000
Interest Amount: Rs. 45000
EMI Amount: Rs. 3781.28
Final Payable Amount: Rs. 92500

e. Display the loan amount, interest, and repayment details clearly.

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C++ PROGRAMMING

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```
1 #include <iostream>
2 using namespace std;
3
4 float calculateEMI(float loanAmount, int years, float rate = 7.5);
5 inline float calculateInterest(float amount, float rate, int years);
6 void updatePayable(float Amount, float subsidy);
7
8 int main()
9 {
10     float loanAmount;
11     float rate;
12     float emi;
13     float interest;
14     float payableAmount;
15     float subsidy;
16     int years;
17
18     cout << "Loan Repayment System" << endl;
19     cout << "Enter Loan Amount: ";
20     cin >> loanAmount;
```

OUTPUT

```
Loan Repayment System
Enter Loan Amount: Enter Loan Period (Years): Enter Interest Rate (Enter 0 for
default): Enter Subsidy Amount:
Loan Amount: Rs. 300000
Interest Amount: Rs. 105000
EMI Amount: Rs. 1779.49
Final Payable Amount: Rs. 405000
```

300000
5
7
0

4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System.

a. Create a class Parcel with appropriate private data members such as parcel type, weight, distance, and packaging charge.

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```
1 #include <iostream>
2 using namespace std;
3
4 class Parcel
5 {
6 private:
7     string parcelType;
8     float weight;
9     float distance;
10    float packagingCharge;
11
12 public:
13
14    void getDetails(float packCharge = 50)
15    {
16        cout << "Enter Parcel Type (Normal/Express): ";
17        cin >> parcelType;
18
19        cout << "Enter Parcel Weight (kg): ";
20        cin >> weight;
```

OUTPUT

```
Enter Parcel Type (Normal/Express): Enter Parcel Weight (kg): Enter Delivery
Distance (km):
Parcel Type : Normal
Weight : 5 kg
Distance : 20 km
Packaging Charge : Rs. 50
Total Delivery Charge : Rs. 250
```

Normal
5
20

b. Use public member functions to read parcel details, calculate the delivery charge, and display the parcel information.

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C++ PROGRAMMING

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1924251018A

```

1 #include <iostream>
2 using namespace std;
3
4 class Parcel
5 {
6 private:
7     string parcelType;
8     float weight;
9     float distance;
10    float packagingCharge;
11
12 public:
13
14    void getDetails(float packCharge = 50)
15    {
16        cout << "Enter Parcel Type (Normal/Express): ";
17        cin >> parcelType;
18
19        cout << "Enter Parcel Weight (kg): ";
20        cin >> weight;
21    }
22
23 };
24
25 int main()
26 {
27     Parcel p;
28     p.getDetails();
29     return 0;
30 }

```

Express
5
20

OUTPUT

```

Enter Parcel Type (Normal/Express): Enter Parcel Weight (kg): Enter Delivery
Distance (km):
Parcel Type : Express
Weight : 5 kg
Distance : 20 km
Packaging Charge : Rs. 50
Total Delivery Charge : Rs. 360

```

c. Use function overloading to calculate charges for Normal Delivery and Express Delivery.

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C++ PROGRAMMING

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1924251018A

```

1 #include <iostream>
2 using namespace std;
3
4 class Parcel
5 {
6 private:
7     string parcelType;
8     float weight;
9     float distance;
10    float packagingCharge;
11
12 public:
13
14    void getDetails(float packCharge = 50)
15    {
16        cout << "Enter Parcel Type (Normal/Express): ";
17        cin >> parcelType;
18
19        cout << "Enter Parcel Weight (kg): ";
20        cin >> weight;
21    }
22
23 };
24
25 int main()
26 {
27     Parcel p;
28     p.getDetails();
29     return 0;
30 }

```

Express
8
30

OUTPUT

```

Enter Parcel Type (Normal/Express): Enter Parcel Weight (kg): Enter Delivery
Distance (km):
Parcel Type : Express
Weight : 8 kg
Distance : 30 km
Packaging Charge : Rs. 50
Total Delivery Charge : Rs. 530

```

d. Use default arguments to assign a default packaging charge when it is not specified by the user.

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C++ PROGRAMMING

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```

1 #include <iostream>
2 using namespace std;
3
4 class Parcel
5 {
6 private:
7     string parcelType;
8     float weight;
9     float distance;
10    float packagingCharge;
11
12 public:
13
14    void getDetails(float packCharge = 50)
15    {
16        cout << "Enter Parcel Type (Normal/Express): ";
17        cin >> parcelType;
18
19        cout << "Enter Parcel Weight (kg): ";
20        cin >> weight;
21    }
22
23 };
24
25 int main()
26 {
27     Parcel p;
28     p.getDetails();
29     return 0;
30 }

```

Normal
4
10

OUTPUT

```

Enter Parcel Type (Normal/Express): Enter Parcel Weight (kg): Enter Delivery
Distance (km):
Parcel Type : Normal
Weight : 4 kg
Distance : 10 km
Packaging Charge : Rs. 50
Total Delivery Charge : Rs. 50

```

e. Use an inline function to calculate the base charge based on parcel weight and delivery distance.

```

1 #include <iostream>
2 using namespace std;
3
4 class Parcel
5 {
6 private:
7     string parcelType;
8     float weight;
9     float distance;
10    float packagingCharge;
11
12 public:
13
14    void getDetails(float packCharge = 50)
15    {
16        cout << "Enter Parcel Type (Normal/Express): ";
17        cin >> parcelType;
18
19        cout << "Enter Parcel Weight (kg): ";
20        cin >> weight;
21    }
22
23
24
25
26
27

```

OUTPUT

```

Enter Parcel Type (Normal/Express): Enter Parcel Weight (kg): Enter Delivery
Distance (km):
Parcel Type : Normal
Weight : 6 kg
Distance : 10 km
Packaging Charge : Rs. 50
Total Delivery Charge : Rs. 220

```

f. Maintain the total number of parcels processed using a static data member and display it through a static member function.

```

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1 #include <iostream>
2 using namespace std;
3
4 class Parcel
5 {
6 private:
7     string parcelType;
8     float weight;
9     float distance;
10    float packagingCharge;
11
12 public:
13
14    void getDetails(float packCharge = 50)
15    {
16        cout << "Enter Parcel Type (Normal/Express): ";
17        cin >> parcelType;
18
19        cout << "Enter Parcel Weight (kg): ";
20        cin >> weight;
21    }
22
23
24
25
26
27

```

OUTPUT

```

Enter Parcel Type (Normal/Express): Enter Parcel Weight (kg): Enter Delivery
Distance (km):
Parcel Type : Normal
Weight : 5 kg
Distance : 20 km
Packaging Charge : Rs. 50
Total Delivery Charge : Rs. 250

```

5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Water Billing System. create a class Consumer with appropriate private data members such as consumer ID, consumer type, water consumption (in liters), and bill amount.

b. Use public member functions to input consumer details, calculate the water bill, and display the billing information.

```

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1 #include <iostream>
2 using namespace std;
3
4 class Consumer
5 {
6 private:
7     int consumerID;
8     string consumerType;
9     float waterConsumption;
10    float billAmount;
11
12 public:
13
14    static int totalConsumers;
15
16    void getDetails();
17    float calculateBill(float liters);
18    float calculateBill(float liters, int commercial);
19    void applySubsidy(float bill, float subsidy);
20    void addServiceTax(float tax = 5);
21    void display();
22    static void calculateTotalConsumption();
23
24
25
26
27
28

```

OUTPUT

```

Consumption (liters): Enter Government Subsidy:
----- Consumer Billing Details -----
Consumer ID : 101
Consumer Type : Domestic
Water Consumption : 500 liters
Final Bill Amount : Rs. 241.5
Total Consumers Processed : 1

```

c. Implement function overloading to calculate water bills for Domestic and Commercial consumers using different tariff rates.

```

#include <iostream>
using namespace std;

class Consumer
{
private:
    int consumerID;
    string consumerType;
    float waterConsumption;
    float billAmount;
public:
    static int totalConsumers;

    void getDetails();
    float calculateBill(float liters);
    float calculateBill(float liters, int commercial);
    void applySubsidy(float bill, float subsidy);
    void addServiceTax(float tax = 5);
    void display();
    static void showTotalConsumers();
};

int totalConsumers = 1;
int consumerID = 102;
string consumerType = "Commercial";
float waterConsumption = 600;
float billAmount = 30;

void Consumer::getDetails()
{
    cout << "Consumer ID : " << consumerID << endl;
    cout << "Consumer Type : " << consumerType << endl;
    cout << "Water Consumption : " << waterConsumption << " liters" << endl;
    cout << "Final Bill Amount : Rs. " << billAmount << endl;
}

void Consumer::calculateBill(float liters)
{
    billAmount = liters * 0.75;
}

void Consumer::calculateBill(float liters, int commercial)
{
    billAmount = liters * 0.75 * commercial;
}

void Consumer::applySubsidy(float bill, float subsidy)
{
    billAmount = bill - subsidy;
}

void Consumer::addServiceTax(float tax)
{
    billAmount = billAmount * (1 + tax / 100);
}

void Consumer::display()
{
    getDetails();
}

void Consumer::showTotalConsumers()
{
    cout << "Total Consumers Processed : " << totalConsumers << endl;
}

int main()
{
    Consumer c;
    c.getDetails();
    c.calculateBill(600);
    c.applySubsidy(472.5, 50);
    c.addServiceTax(5);
    c.display();
    c.showTotalConsumers();
}

```

OUTPUT

```

Consumption (liters): Enter Government Subsidy:
----- Consumer Billing Details -----
Consumer ID : 102
Consumer Type : Commercial
Water Consumption : 600 liters
Final Bill Amount : Rs. 472.5
Total Consumers Processed : 1

```

d. Use call by reference to update the bill amount after applying a government subsidy.

```

#include <iostream>
using namespace std;

class Consumer
{
private:
    int consumerID;
    string consumerType;
    float waterConsumption;
    float billAmount;
public:
    static int totalConsumers;

    void getDetails();
    float calculateBill(float liters);
    float calculateBill(float liters, int commercial);
    void applySubsidy(float bill, float subsidy);
    void addServiceTax(float tax = 5);
    void display();
    static void showTotalConsumers();
};

int totalConsumers = 1;
int consumerID = 103;
string consumerType = "Domestic";
float waterConsumption = 1000;
float billAmount = 50;

void Consumer::getDetails()
{
    cout << "Consumer ID : " << consumerID << endl;
    cout << "Consumer Type : " << consumerType << endl;
    cout << "Water Consumption : " << waterConsumption << " liters" << endl;
    cout << "Final Bill Amount : Rs. " << billAmount << endl;
}

void Consumer::calculateBill(float liters)
{
    billAmount = liters * 0.75;
}

void Consumer::calculateBill(float liters, int commercial)
{
    billAmount = liters * 0.75 * commercial;
}

void Consumer::applySubsidy(float bill, float subsidy)
{
    billAmount = bill - subsidy;
}

void Consumer::addServiceTax(float tax)
{
    billAmount = billAmount * (1 + tax / 100);
}

void Consumer::display()
{
    getDetails();
}

void Consumer::showTotalConsumers()
{
    cout << "Total Consumers Processed : " << totalConsumers << endl;
}

int main()
{
    Consumer c;
    c.getDetails();
    c.calculateBill(1000);
    c.applySubsidy(472.5, 50);
    c.addServiceTax(5);
    c.display();
    c.showTotalConsumers();
}

```

OUTPUT

```

Enter Number of Consumers:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (liters): Enter Government Subsidy:
----- Consumer Billing Details -----
Consumer ID : 103
Consumer Type : Domestic
Water Consumption : 1000 liters
Final Bill Amount : Rs. 472.5
Total Consumers Processed : 1

```

e. Use default arguments to include a service tax, where a default tax rate is applied unless specified.

```

#include <iostream>
using namespace std;

class Consumer
{
private:
    int consumerID;
    string consumerType;
    float waterConsumption;
    float billAmount;
public:
    static int totalConsumers;

    void getDetails();
    float calculateBill(float liters);
    float calculateBill(float liters, int commercial);
    void applySubsidy(float bill, float subsidy);
    void addServiceTax(float tax = 5);
    void display();
    static void showTotalConsumers();
};

int totalConsumers = 1;
int consumerID = 104;
string consumerType = "Commercial";
float waterConsumption = 1200;
float billAmount = 100;

void Consumer::getDetails()
{
    cout << "Consumer ID : " << consumerID << endl;
    cout << "Consumer Type : " << consumerType << endl;
    cout << "Water Consumption : " << waterConsumption << " liters" << endl;
    cout << "Final Bill Amount : Rs. " << billAmount << endl;
}

void Consumer::calculateBill(float liters)
{
    billAmount = liters * 0.75;
}

void Consumer::calculateBill(float liters, int commercial)
{
    billAmount = liters * 0.75 * commercial;
}

void Consumer::applySubsidy(float bill, float subsidy)
{
    billAmount = bill - subsidy;
}

void Consumer::addServiceTax(float tax)
{
    billAmount = billAmount * (1 + tax / 100);
}

void Consumer::display()
{
    getDetails();
}

void Consumer::showTotalConsumers()
{
    cout << "Total Consumers Processed : " << totalConsumers << endl;
}

int main()
{
    Consumer c;
    c.getDetails();
    c.calculateBill(1200);
    c.applySubsidy(903, 50);
    c.addServiceTax(5);
    c.display();
    c.showTotalConsumers();
}

```

OUTPUT

```

Enter Number of Consumers:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (liters): Enter Government Subsidy:
----- Consumer Billing Details -----
Consumer ID : 104
Consumer Type : Commercial
Water Consumption : 1200 liters
Final Bill Amount : Rs. 903
Total Consumers Processed : 1

```

f. Declare all functions using function prototyping before their definitions.

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```

1 #include <iostream>
2 using namespace std;
3
4 class Consumer
5 {
6 private:
7     int consumerID;
8     string consumerType;
9     float waterConsumption;
10    float billAmount;
11
12 public:
13     static int totalConsumers;
14
15     void getDetails();
16     float calculateBill(float liters);
17     float calculateBill(float liters, int commercial);
18     void applySubsidy(float &bill, float subsidy);
19     void addServiceTax(float tax = 5);
20     void display();
21     static void showTotalConsumers();

```

```

1
105
Industrial
700
20

```

OUTPUT

```

Enter Number of Consumers:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Invalid Consumer Type
Enter Government Subsidy:
----- Consumer Billing Details -----
Consumer ID : 105
Consumer Type : Industrial
Water Consumption : 700 Liters

```

g. Maintain the total number of consumers processed using a static data member and display it through a static member function.

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1924251618A

```

1 #include <iostream>
2 using namespace std;
3
4 class Consumer
5 {
6 private:
7     int consumerID;
8     string consumerType;
9     float waterConsumption;
10    float billAmount;
11
12 public:
13     static int totalConsumers;
14
15     void getDetails();
16     float calculateBill(float liters);
17     float calculateBill(float liters, int commercial);
18     void applySubsidy(float &bill, float subsidy);
19     void addServiceTax(float tax = 5);
20     void display();
21     static void showTotalConsumers();

```

```

3
101
Domestic
500
20
102
Commercial
600
30
103
Domestic
800
40

```

OUTPUT

```

Enter Number of Consumers:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Enter Government Subsidy:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Enter Government Subsidy:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Enter Government Subsidy:
----- Consumer Billing Details -----

```

h. Create an array of objects to store and process the details of multiple consumers, and display the consumer type, water consumption, subsidy-adjusted bill amount,



```
1 #include <iostream>
2 using namespace std;
3
4 class Consumer
5 {
6 private:
7     int consumerID;
8     string consumerType;
9     float waterConsumption;
10    float billAmount;
11
12 public:
13     static int totalConsumers;
14
15     void getDetails();
16     float calculateBill(float liters);
17     float calculateBill(float liters, int commercial);
18     void applySubsidy(float &bill, float subsidy);
19     void addServiceTax(float tax = 5);
20     void display();
21     static void showTotalConsumers();
```

OUTPUT

```
Enter Number of Consumers:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Enter Government Subsidy:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Enter Government Subsidy:
Enter Consumer ID: Enter Consumer Type (Domestic/Commercial): Enter Water
Consumption (Liters): Enter Government Subsidy:
----- Consumer Billing Details -----
```

```
3
101
Domestic
500
20
102
Commercial
600
30
103
Domestic
800
40
```